NTRK1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes NTRK1 gene fusions from the msk_impact_50k_2026 study, covering 78 fusion events. 32/78 fusions (41.0%) were in-frame. 64/78 fusions (82.1%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test (p=0.0482628, statistically significant at alpha=0.05). No literature benchmark is configured for NTRK1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 78 analyzed fusion events, identifying 43 distinct fusion partner genes. The most frequent partner was LMNA, observed in 12 events.

Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 30/32 (93.8%) of in-frame fusions retained the domain, p=0.0482628 (statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

Domain disruption

Domain-disruption analysis was skipped: NTRK1 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 76 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 501 aa (permutation-corrected p=0.014985, statistically significant at alpha=0.05). This is -11 aa from the nearest configured domain boundary at 512 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=64) and "not_retained" (n=12). For Is_protein_fusion, retained: 64/64 (100.0%, 95% CI 94.3%-100.0%); not_retained: 12/12 (100.0%, 95% CI 75.8%-100.0%). Welch's t-test comparing Tumor_variant_count between groups gave means of 66.3594 vs 43.25, p=0.200314 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

Joint partner

Joint partner reported: NTRK1 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
LMNA	12	0.15384615384615385	0.30004340774793187	None	0.851937984496124	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.41749936555355677	1
TPM3	12	0.15384615384615385	0.30004340774793187	None	0.7467054263565891	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.397768260902394	2
MEF2D	1	0.01282051282051282	0.30004340774793187	None	1.0	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3923763780759181	3
TRPM8	1	0.01282051282051282	0.30004340774793187	None	1.0	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3923763780759181	4
ARHGEF11	2	0.02564102564102564	0.30004340774793187	None	0.9430232558139535	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.38650093084872666	5
RABGAP1L	1	0.01282051282051282	0.30004340774793187	None	0.8860465116279069	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3710100990061506	6
VANGL2	1	0.01282051282051282	0.30004340774793187	None	0.8860465116279069	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3710100990061506	7
ZBTB7B	1	0.01282051282051282	0.30004340774793187	None	0.8860465116279069	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3710100990061506	8
SCP2	1	0.01282051282051282	0.30004340774793187	None	0.8651162790697674	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.36708568040149947	9
TPR	5	0.0641025641025641	0.30004340774793187	None	0.7427906976744186	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.36338040312064085	10
LTAP1	1	0.01282051282051282	0.30004340774793187	None	0.8372093023255813	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3618531222619646	11

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ATP1A2	1	0.01282051282051282	0.30004340774793187	None	0.8186046511627907	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.35836475016894137	12
PLEKHA6	3	0.038461538461538464	0.30004340774793187	None	0.7627906976744185	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3575150185052562	13
IRF2BP2	3	0.038461538461538464	0.30004340774793187	None	0.758139534883721	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3566429254820004	14
ANKRD36	1	0.01282051282051282	0.30004340774793187	None	0.8069767441860465	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.35618451761080183	15
RAB25	1	0.01282051282051282	0.30004340774793187	None	0.8046511627906977	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3557484710991739	16
IQGAP3	1	0.01282051282051282	0.30004340774793187	None	0.7976744186046512	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3544403315642902	17
SMYD2	1	0.01282051282051282	0.30004340774793187	None	0.7767441860465116	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.350515912959639	18
EML4	1	0.01282051282051282	0.30004340774793187	None	0.7627906976744185	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3478996338898716	19
EPS15	1	0.01282051282051282	0.30004340774793187	None	0.7627906976744185	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3478996338898716	20
F11R	1	0.01282051282051282	0.30004340774793187	None	0.7627906976744185	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3478996338898716	21
PEAR1	1	0.01282051282051282	0.30004340774793187	None	0.7627906976744185	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3478996338898716	22
AFAP1	1	0.01282051282051282	0.30004340774793187	None	0.7488372093023256	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.34528335482010414	23
DIAPH1	1	0.01282051282051282	0.30004340774793187	None	0.7488372093023256	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.34528335482010414	24

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TRIM63	1	0.01282051282051282	0.30004340774793187	None	0.7488372093023256	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.34528335482010414	25
STK11	1	0.01282051282051282	0.30004340774793187	None	0.7325581395348837	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3422310292387088	26
KIF21B	1	0.01282051282051282	0.30004340774793187	None	0.7255813953488373	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.34092288970382506	27
GON4L	1	0.01282051282051282	0.30004340774793187	None	0.7162790697674419	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3391787036573135	28
BCAN	2	0.02564102564102564	0.30004340774793187	None	0.49534883720930234	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.30256197736035456	29
CTRC	2	0.02564102564102564	0.30004340774793187	None	0.49534883720930234	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.30256197736035456	30
NOS1AP	1	0.01282051282051282	0.30004340774793187	None	0.5186046511627906	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3021147501689414	31
NELFCD	1	0.01282051282051282	0.30004340774793187	None	0.49534883720930234	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.29775428505266227	32
SEMA4A	1	0.01282051282051282	0.30004340774793187	None	0.49534883720930234	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.29775428505266227	33
COP1	1	0.01282051282051282	0.30004340774793187	None	0.4534883720930233	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.28990544784336	34
TARS2	1	0.01282051282051282	0.30004340774793187	None	0.2697674418604651	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.2554577734247553	35
P2RY8	1	0.01282051282051282	0.30004340774793187	None	0.2581395348837209	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.2532775408666158	36
DDR2	1	0.01282051282051282	0.30004340774793187	None	None	0.8504409667759769	['confidence_certainty', 'domain_retention', 'recurrence']	0.25215554224728387	37

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
METTL25B	1	0.01282051282051282	0.30004340774793187	None	None	0.8504409667759769	['confidence_certainty', 'domain_retention', 'recurrence']	0.25215554224728387	38
SLAMF6	3	0.038461538461538464	0.30004340774793187	None	0.11395348837209307	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.2358580417610702	39
CADM3	1	0.01282051282051282	0.30004340774793187	None	0.11395348837209307	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.22624265714568556	40
PRCC	1	0.01282051282051282	0.30004340774793187	None	0.11395348837209307	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.22624265714568556	41
TAFA2	1	0.01282051282051282	0.30004340774793187	None	0.11395348837209307	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.22624265714568556	42
SHPRH	1	0.01282051282051282	0.30004340774793187	None	0.0	0.8504409667759769	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.20487637807591816	43

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
71	0	64	1	11	0.0	0.15789473684210525	0.8016323462331666
120	5	59	2	10	0.423728813559322	0.3043106841520683	0.5166827995746912
182	5	59	3	9	0.2542372881355932	0.10748982443269033	0.9686326464637113
187	6	58	3	9	0.3103448275862069	0.14608214986786255	0.83540284831964
284	10	54	5	7	0.25925925925925924	0.05253839246726117	1.2795232195834179
294	11	53	5	7	0.29056603773584905	0.11464025310482523	0.9406628637664377
321	13	51	5	7	0.3568627450980392	0.14136093824402443	0.8496705809825442
343	14	50	5	7	0.392	0.1614921032975089	0.7918487091109843
359	15	49	5	7	0.42857142857142855	0.28166616270038797	0.550265322823117
377	16	48	5	7	0.4666666666666667	0.29450254706111656	0.530910944782467
379	17	47	5	7	0.5063829787234042	0.31247043251825074	0.505191071445305
383	18	46	5	7	0.5478260869565217	0.494013437398657	0.30626123790080845
393	27	37	5	7	1.0216216216216216	1.0	-0.0

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
399	39	25	5	7	2.184	0.33960073354007925	0.46903138034780045
405	39	25	6	6	1.56	0.5328211446391313	0.27341854881920497
411	40	24	6	6	1.6666666666666667	0.5236156499522178	0.28098738162793063
414	41	23	6	6	1.7826086956521738	0.5184932375861125	0.2852569034951671
417	46	18	7	5	1.8253968253968254	0.49401343739865705	0.3062612379008084
418	48	16	7	5	2.142857142857143	0.29450254706111656	0.530910944782467
423	49	15	7	5	2.3333333333333335	0.28166616270038797	0.550265322823117
443	50	14	7	5	2.5510204081632653	0.1614921032975089	0.7918487091109843
452	61	3	8	4	10.166666666666666	0.010191376831302902	1.9917671398557228
492	62	2	8	4	15.5	0.004800736843751964	2.3186920995794114
501	64	0	9	3	None	0.0031294452347083927	2.5045326441976177
571	64	0	10	2	None	0.023157894736842103	1.6353009244666417
616	64	0	11	1	None	0.15789473684210525	0.8016323462331666

domain_retention tables

frame_domain_contingency_table

30	34
2	10

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	76	64	3	9	0.1117283950617284

frequency tables

Partner_gene_counts

Partner_gene	Event_count
AFAP1	1
ANKRD36	1
ARHGEF11	2
ATP1A2	1
BCAN	2
CADM3	1

Partner_gene	Event_count
COP1	1
CTRC	2
DDR2	1
DIAPH1	1
EML4	1
EPS15	1
F11R	1
GON4L	1
IQGAP3	1
IRF2BP2	3
KIF21B	1
LMNA	12
LTAP1	1
MEF2D	1
METTL25B	1
NELFCD	1
NOS1AP	1
P2RY8	1
PEAR1	1
PLEKHA6	3
PRCC	1
RAB25	1
RABGAP1L	1
SCP2	1
SEMA4A	1
SHPRH	1
SLAMF6	3
SMYD2	1
STK11	1
TAFA2	1
TARS2	1
TPM3	12
TPR	5
TRIM63	1
TRPM8	1

Partner_gene	Event_count
VANGL2	1
ZBTB7B	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013559-T01-IM5-1	P-0013559-T01-IM5		AFAP1-NTRK1	AFAP1	in-frame	three_prime	156844093	9	393	True	retained	1.0	False	Protein kinase domain			AFAP1(NM_001134647) - NTRK1(NM_002529) Rearrangement : t(1;4)(q23.1;p16.1)(chr1:g.156844093::chr4:g.7852375) | Protein fusion: in frame (AFAP1-NTRK1)	NA
EVT-P-0042270-T01-IM6-2	P-0042270-T01-IM6		ANKRD36-NTRK1	ANKRD36	out-of-frame	three_prime	156844593	11	418	True	retained	1.0	False	Protein kinase domain			ANKRD36 (NM_001164315) - NTRK1 (NM_002529) fusion: t(1;2)(q23.1;q11.2)(chr1:g.156844593::chr2:g.97835462) | Protein Fusion: out of frame {ANKRD36:NTRK1}	NA
EVT-P-0017558-T02-IM7-3	P-0017558-T02-IM7		ARHGEF11-NTRK1	ARHGEF11	out-of-frame	three_prime	156845502	12	501	True	retained	1.0	False	Protein kinase domain			ARHGEF11 (NM_198236) - NTRK1 (NM_002529) fusion: c.32+488:ARHGEF11_c.1501+44:NTRK1inv | Protein Fusion: out of frame {ARHGEF11:NTRK1}	NA
EVT-P-0050637-T01-IM6-4	P-0050637-T01-IM6		BCAN-NTRK1	BCAN	in-frame	three_prime	156842373	7	284	True	retained	1.0	False	Protein kinase domain			BCAN (NM_021948) - NTRK1 (NM_002529) fusion: c.1064-525:BCAN_c.850+826:NTRK1del | Protein Fusion: in frame {BCAN:NTRK1}	NA
EVT-P-0041874-T01-IM6-5	P-0041874-T01-IM6		BCAN-NTRK1	BCAN	in-frame	three_prime	156843258	8	284	True	retained	1.0	False	Protein kinase domain			BCAN (NM_021948) - NTRK1 (NM_002529) fusion: c.92-224:BCAN_c.851-167:NTRK1del | Protein Fusion: in frame {BCAN:NTRK1}	NA
EVT-P-0055952-T01-IM6-6	P-0055952-T01-IM6		COP1-NTRK1	COP1	in-frame	three_prime	156851106	17	736	True	disrupted	0.17037037037037037	True			Protein kinase domain	RFWD2 (NM_022457) - NTRK1 (NM_002529) rearrangement: c.2133+18462:RFWD2_c.2206-143:NTRK1inv | Protein Fusion: in frame {RFWD2:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014859-T03-IM6-8	P-0014859-T03-IM6		CTRC-NTRK1	CTRC	unknown	three_prime	156841593	7	284	True	retained	1.0	False	Protein kinase domain			CTRC (NM_007272) - NTRK1 (NM_002529) rearrangement: c.88:CTRC_c.850+46:NTRK1del | Protein Fusion: mid-exon {CTRC:NTRK1}	NA
EVT-P-0014859-T01-IM6-9	P-0014859-T01-IM6		CTRC-NTRK1	CTRC	unknown	three_prime	156841592	7	284	True	retained	1.0	False	Protein kinase domain			CTRC (NM_007272) - NTRK1 (NM_002529) rearrangement: c.88:CTRC_c.850+45:NTRK1del | Protein fusion: mid-exon (CTRC-NTRK1)	NA
EVT-P-0029326-T01-IM6-10	P-0029326-T01-IM6		DIAPH1-NTRK1	DIAPH1	in-frame	three_prime	156843860	8	393	True	retained	1.0	False	Protein kinase domain			DIAPH1 (NM_005219) - NTRK1 (NM_002529) rearrangement: t(1;5)(q23.1;q31.3)(chr1:g.156843860::chr5:g.140904837) | Protein Fusion: in frame {DIAPH1:NTRK1}	NA
EVT-P-0059139-T01-IM7-11	P-0059139-T01-IM7		EML4-NTRK1	EML4	in-frame	three_prime	156844256	9	399	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - NTRK1 (NM_002529) fusion: t(1;2)(q23.1;p21)(chr1:g.156844256::chr2:g.42473987) | Protein Fusion: in frame {EML4:NTRK1}	NA
EVT-P-0037901-T01-IM6-12	P-0037901-T01-IM6		EPS15-NTRK1	EPS15	in-frame	three_prime	156844212	9	399	True	retained	1.0	False	Protein kinase domain			EPS15 (NM_001981) - NTRK1 (NM_002529) rearrangement: c.2119+1240:EPS15_c.1195+20:NTRK1inv | Protein Fusion: in frame {EPS15:NTRK1}	NA
EVT-P-0009855-T01-IM5-13	P-0009855-T01-IM5		GON4L-NTRK1	GON4L	unknown	three_prime	156843711	8	379	False	retained	1.0	False	Protein kinase domain			GON4L (NM_001037533) - NTRK1 (NM_002529) rearrangement : c.4141:GON4L_c.1137:NTRK1inv | Protein fusion: mid-exon (GON4L-NTRK1)	NA
EVT-P-0050669-T01-IM6-15	P-0050669-T01-IM6		IQGAP3-NTRK1	IQGAP3	unknown	three_prime	156844408	10	414	False	retained	1.0	False	Protein kinase domain			IQGAP3 (NM_178229) - NTRK1 (NM_002529) fusion: c.2531-131:IQGAP3_c.1241:NTRK1inv | Protein Fusion: mid-exon {IQGAP3:NTRK1}	NA
EVT-P-0022925-T01-IM6-16	P-0022925-T01-IM6		LMNA-NTRK1	LMNA	unknown	three_prime	156844398	10	411	False	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-3 fused to NTRK1 exons 10 - 17): c.513+1675:LMNA_c.1231:NTRK1del | Protein Fusion: mid-exon {LMNA:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010581-T01-IM5-17	P-0010581-T01-IM5		LMNA-NTRK1	LMNA	in-frame	three_prime	156844556	10	417	True	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK (NM_002529) fusion (LMNA exons 1-2 fused to NTRK1 exons 11-17): c.514-811:LMNA_c.1251+138:NTRK1del | Protein fusion: in frame (LMNA-NTRK1)	NA
EVT-P-0010086-T01-IM5-18	P-0010086-T01-IM5		LMNA-NTRK1	LMNA	in-frame	three_prime	156844436	10	417	True	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) rearrangement: c.514-1686:LMNA_c.1251+18:NTRK1del | Protein fusion: in frame (LMNA-NTRK1)	NA
EVT-P-0007739-T01-IM5-19	P-0007739-T01-IM5		LMNA-NTRK1	LMNA	unknown	three_prime	156845035	11	452	True	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) Fusion (LMNA exons 1-8 fused to NTRK1 exons 12-17): c.1443:LMNA_c.1354+235:NTRK1 | Protein fusion: mid-exon (LMNA-NTRK1)	NA
EVT-P-0003705-T01-IM5-24	P-0003705-T01-IM5		LMNA-NTRK1	LMNA	unknown	three_prime	156845433	12	492	False	retained	1.0	False	Protein kinase domain			c.686:LMNA_c.1476:NTRK1del | Protein fusion: mid-exon (LMNA-NTRK1)	NA
EVT-P-0003329-T04-IM6-25	P-0003329-T04-IM6		LMNA-NTRK1	LMNA	unknown	three_prime	156845200	12	452	True	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-2 fused to NTRK1 exons 12 - 17): c.495:LMNA_c.1355-112:NTRK1del | Protein Fusion: mid-exon {LMNA:NTRK1}	NA
EVT-P-0003329-T03-IM6-26	P-0003329-T03-IM6		LMNA-NTRK1	LMNA	unknown	three_prime	156845200	12	452	True	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-2 fused to NTRK1 exons 12 - 17): c.495:LMNA_c.1355-112:NTRK1del | Protein Fusion: mid-exon {LMNA:NTRK1}	NA
EVT-P-0066239-T01-IM7-27	P-0066239-T01-IM7		LMNA-NTRK1	LMNA	unknown	three_prime	156845170	12	452	True	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1428:LMNA_c.1355-142:NTRK1del | Protein Fusion: mid-exon {LMNA:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003329-T02-IM6-28	P-0003329-T02-IM6		LMNA-NTRK1	LMNA	unknown	three_prime	156845200	12	452	True	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) Fusion (LMNA exon2 fused with NTRK1 exon12): c.495:LMNA_c.1355-112:NTRK1del | Protein Fusion: mid-exon {LMNA:NTRK1}	NA
EVT-P-0023994-T06-IM6-29	P-0023994-T06-IM6		LMNA-NTRK1	LMNA	in-frame	three_prime	156844525	10	417	True	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1699-362:LMNA_c.1251+107:NTRK1del | Protein Fusion: in frame {LMNA:NTRK1}	NA
EVT-P-0023994-T05-IM6-30	P-0023994-T05-IM6		LMNA-NTRK1	LMNA	in-frame	three_prime	156844525	10	417	True	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) Fusion (LMNA exon 10 fused with NTRK1 exon 11) : c.1699-362:LMNA_c.1251+107:NTRK1del | Protein Fusion: in frame {LMNA:NTRK1}	NA
EVT-P-0023994-T01-IM6-31	P-0023994-T01-IM6		LMNA-NTRK1	LMNA	in-frame	three_prime	156844525	10	417	True	retained	1.0	False	Protein kinase domain			LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-10 fused to NTRK1 exons 11-17): c.1699-362:LMNA_c.1251+107:NTRK1del | Protein Fusion: in frame {LMNA:NTRK1}	NA
EVT-P-0025031-T02-IM6-32	P-0025031-T02-IM6		LTAP1-NTRK1	LTAP1	unknown	three_prime	156846272	14	571	False	disrupted	0.78148148148148144	True			Protein kinase domain	C1orf43 (NM_001098616) - NTRK1 (NM_002529): c.168+114:C1orf43_c.1713:NTRK1inv | Protein Fusion: mid-exon {C1orf43:NTRK1}	NA
EVT-P-0024231-T01-IM6-33	P-0024231-T01-IM6		MEF2D-NTRK1	MEF2D	in-frame	five_prime	156845756	13	501	True	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - MEF2D (NM_005920) rearrangement: c.1502-116:NTRK1_c.1006+1908:MEF2Binv | Protein Fusion: in frame {NTRK1:MEF2D}	NA
EVT-P-0045731-T01-IM6-35	P-0045731-T01-IM6		NOS1AP-NTRK1	NOS1AP	unknown	three_prime	156843455	8	294	False	retained	1.0	False	Protein kinase domain			NOS1AP (NM_014697) - NTRK1 (NM_002529) fusion: c.1106-363:NOS1AP_c.881:NTRK1dup | Protein Fusion: mid-exon {NOS1AP:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026354-T01-IM6-36	P-0026354-T01-IM6		NTRK1-ARHGEF11	ARHGEF11	in-frame	three_prime	156845184	12	452	True	retained	1.0	False	Protein kinase domain			ARHGEF11 (NM_198236) - NTRK1 (NM_002529) rearrangement: c.4631-285:ARHGEF11_c.1355-128:NTRK1inv | Protein Fusion: in frame {ARHGEF11:NTRK1}	NA
EVT-P-0035466-T01-IM6-37	P-0035466-T01-IM6		NTRK1-ATP1A2	ATP1A2	unknown	three_prime	156844714	11	423	False	retained	1.0	False	Protein kinase domain			ATP1A2 (NM_000702) - NTRK1 (NM_002529) rearrangement: c.2563+2:ATP1A2_c.1268:NTRK1dup | Protein Fusion: mid-exon {ATP1A2:NTRK1}	NA
EVT-P-0058090-T04-IM7-38	P-0058090-T04-IM7		NTRK1-CADM3	CADM3	out-of-frame	five_prime	156836298	4	120	True	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - CADM3 (NM_021189) rearrangement: c.360-404:NTRK1_c.88+5312:CADM3del | Protein Fusion: out of frame {NTRK1:CADM3}	NA
EVT-P-0021262-T01-IM6-41	P-0021262-T01-IM6		NTRK1-F11R	F11R	in-frame	three_prime	156844288	10	399	True	retained	1.0	False	Protein kinase domain			F11R (NM_016946) - NTRK1 (NM_002529) rearrangement: c.389-101:F11R_c.1196-75:NTRK1inv | Protein Fusion: in frame {F11R:NTRK1}	NA
EVT-P-0022467-T01-IM6-42	P-0022467-T01-IM6		NTRK1-IRF2BP2	IRF2BP2	in-frame	three_prime	156844129	9	393	True	retained	1.0	False	Protein kinase domain			IRF2BP2 (NM_001077397) - NTRK1 (NM_002529) fusion (IRF2BP2 exon 1 fused with NTRK1 exons 9-17): c.1001-193:IRF2BP2_c.1178-46:NTRK1inv | Protein Fusion: in frame {IRF2BP2:NTRK1}	NA
EVT-P-0010628-T01-IM5-43	P-0010628-T01-IM5		NTRK1-IRF2BP2	IRF2BP2	unknown	three_prime	156844224	9	399	True	retained	1.0	False	Protein kinase domain			IRF2BP2 (NM_001077397) - NTRK1 (NM_002529) rearrangement : c.948:IRF2BP2_1195+32:NTRK1inv | Protein fusion: mid-exon (IRF2BP2-NTRK1)	NA
EVT-P-0010628-T03-IM6-44	P-0010628-T03-IM6		NTRK1-IRF2BP2	IRF2BP2	unknown	three_prime	156844223	9	399	True	retained	1.0	False	Protein kinase domain			IRF2BP2 (NM_001077397) - NTRK1 (NM_002529) rearrangement: c.950:IRF2BP2_c.1195+31:NTRK1inv | Protein Fusion: mid-exon {IRF2BP2:NTRK1}	NA
EVT-P-0027278-T01-IM6-45	P-0027278-T01-IM6		NTRK1-KIF21B	KIF21B	unknown	three_prime	156843723	8	383	False	retained	1.0	False	Protein kinase domain			KIF21B (NM_001252100) - NTRK1 (NM_002529) rearrangement: c.2077+601:KIF21B_c.1149:NTRK1inv | Protein Fusion: mid-exon {KIF21B:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0056197-T01-IM6-46	P-0056197-T01-IM6		NTRK1-NELFCD	NELFCD	out-of-frame	five_prime	156843148	8	284	True	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - NELFCD (NM_198976) rearrangement: t(1;20)(q23.1;q13.32)(chr1:g.156843148::chr20:g.57565762) | Protein Fusion: out of frame {NTRK1:NELFCD}	NA
EVT-P-0027277-T01-IM6-79	P-0027277-T01-IM6		NTRK1-PEAR1	PEAR1	in-frame	three_prime	156844219	9	399	True	retained	1.0	False	Protein kinase domain			PEAR1 (NM_001080471) - NTRK1 (NM_002529) fusion (PEAR1 exons 1-15 fused to NTRK1 exons 9-17): c.1951+466:PEAR1_c.1195+27:NTRK1dup | Protein Fusion: in frame {PEAR1:NTRK1}	NA
EVT-P-0029892-T05-IM6-80	P-0029892-T05-IM6		NTRK1-PLEKHA6	PLEKHA6	in-frame	three_prime	156844239	9	399	True	retained	1.0	False	Protein kinase domain			PLEKHA6 (NM_014935) - NTRK1 (NM_002529) fusion: c.2032+105:PLEKHA6_c.1195+47:NTRK1inv | Protein Fusion: in frame {PLEKHA6:NTRK1}	NA
EVT-P-0029892-T04-IM6-81	P-0029892-T04-IM6		NTRK1-PLEKHA6	PLEKHA6	in-frame	three_prime	156844239	9	399	True	retained	1.0	False	Protein kinase domain			PLEKHA6 (NM_014935) - NTRK1 (NM_002529) fusion: c.2032+105:PLEKHA6_c.1195+47:NTRK1inv | Protein Fusion: in frame {PLEKHA6:NTRK1}	NA
EVT-P-0029892-T03-IM6-82	P-0029892-T03-IM6		NTRK1-PLEKHA6	PLEKHA6	in-frame	three_prime	156844239	9	399	True	retained	1.0	False	Protein kinase domain			PLEKHA6 (NM_014935) - NTRK1 (NM_002529) fusion: c.2032+105:PLEKHA6_c.1195+47:NTRK1inv | Protein Fusion: in frame {PLEKHA6:NTRK1}	NA
EVT-P-0038819-T01-IM6-83	P-0038819-T01-IM6		NTRK1-RAB25	RAB25	out-of-frame	five_prime	156844458	10	417	True	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - RAB25 (NM_020387) Rearrangement : | Protein Fusion: out of frame {NTRK1:RAB25}	NA
EVT-P-0015004-T01-IM6-84	P-0015004-T01-IM6		NTRK1-RABGAP1L	RABGAP1L	out-of-frame	five_prime	156844972	11	452	True	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - RABGAP1L (NM_014857) rearrangement: c.1354+172:NTRK1_c.2340+50594:RABGAP1Ldel | Protein fusion: out of frame (NTRK1-RABGAP1L)	NA
EVT-P-0038568-T01-IM6-85	P-0038568-T01-IM6		NTRK1-SEMA4A	SEMA4A	out-of-frame	five_prime	156841963	7	284	True	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - SEMA4A (NM_001193300) rearrangement: c.850+416:NTRK1_c.301-270:SEMA4Adup | Protein Fusion: out of frame {NTRK1:SEMA4A}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0048896-T01-IM6-86	P-0048896-T01-IM6		NTRK1-SMYD2	SMYD2	unknown	five_prime	156844382	10	405	False	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - SMYD2 (NM_020197) rearrangement: c.1215:NTRK1_c.602+876:SMYD2del | Protein Fusion: mid-exon {NTRK1:SMYD2}	NA
EVT-P-0023987-T01-IM6-88	P-0023987-T01-IM6		NTRK1-TPR	TPR	unknown	three_prime	156844285	10	399	True	retained	1.0	False	Protein kinase domain			TPR (NM_003292) - NTRK1 (NM_002529) rearrangement: c.2902:TPR_c.1196-78:NTRK1inv | Protein Fusion: mid-exon {TPR:NTRK1}	NA
EVT-P-0024549-T03-IM6-89	P-0024549-T03-IM6		NTRK1-TPR	TPR	in-frame	three_prime	156844290	10	399	True	retained	1.0	False	Protein kinase domain			TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+651_c.1196-73inv | Protein Fusion: in frame {TPR:NTRK1}	NA
EVT-P-0007050-T01-IM5-90	P-0007050-T01-IM5		NTRK1-VANGL2	VANGL2	unknown	three_prime	156845141	12	452	True	retained	1.0	False	Protein kinase domain			VANGL2 (NM_020335) - NTRK1(NM_002529) rearrangement : c.-191+4456:VANGL2_c.1355-171:NTRK1dup | Transcript fusion (VANGL2-NTRK1)	NA
EVT-P-0006172-T01-IM5-91	P-0006172-T01-IM5		P2RY8-NTRK1	P2RY8	unknown	five_prime	156838012	5	182	False	lost	0.0	False		Protein kinase domain		P2RY8(NM_178129) - NTRK1 (NM_002529) rearrangement: t(X;1)(chrX:g.1598513::chr1:g.156838012) | Protein fusion: mid-exon (NTRK1-P2RY8)	NA
EVT-P-0042526-T02-IM6-92	P-0042526-T02-IM6		PRCC-NTRK1	PRCC	unknown	three_prime	156834591	3	120	False	retained	1.0	False	Protein kinase domain			PRCC (NM_005973) - NTRK1 (NM_002529) fusion: c.1323+10:PRCC_c.359:NTRK1del | Protein Fusion: mid-exon {PRCC:NTRK1}	NA
EVT-P-0044467-T01-IM6-93	P-0044467-T01-IM6		SCP2-NTRK1	SCP2	unknown	three_prime	156844773	11	443	False	retained	1.0	False	Protein kinase domain			SCP2 (NM_002979) - NTRK1 (NM_002529) fusion: c.1338+1979:SCP2_c.1327:NTRK1del | Protein Fusion: mid-exon {SCP2:NTRK1}	NA
EVT-P-0036662-T01-IM6-94	P-0036662-T01-IM6		SHPRH-NTRK1	SHPRH	out-of-frame	five_prime	156831180	1	71	True	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - SHPRH (NM_001042683) rearrangement: t(1;6)(q231;q24.3)(chr1:g.156831180::chr6:g.146225530) | Protein Fusion: out of frame {NTRK1:SHPRH}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0055621-T05-IM6-95	P-0055621-T05-IM6		SLAMF6-NTRK1	SLAMF6	out-of-frame	three_prime	156836411	4	120	True	retained	1.0	False	Protein kinase domain			SLAMF6 (NM_052931) - NTRK1 (NM_002529) fusion: c.50-1837:SLAMF6_c.360-291:NTRK1inv | Protein Fusion: out of frame {SLAMF6:NTRK1}	NA
EVT-P-0055621-T02-IM6-96	P-0055621-T02-IM6		SLAMF6-NTRK1	SLAMF6	out-of-frame	three_prime	156836411	4	120	True	retained	1.0	False	Protein kinase domain			SLAMF6 (NM_052931) - NTRK1 (NM_002529) rearrangement: c.50-1839:SLAMF6_c.360-291:NTRK1inv | Protein Fusion: out of frame {SLAMF6:NTRK1}	NA
EVT-P-0055621-T01-IM6-97	P-0055621-T01-IM6		SLAMF6-NTRK1	SLAMF6	out-of-frame	three_prime	156836411	4	120	True	retained	1.0	False	Protein kinase domain			SLAMF6 (NM_052931) - NTRK1 (NM_002529) rearrangement: c.50-1839:SLAMF6_c.360-291:NTRK1inv | Protein Fusion: out of frame {SLAMF6:NTRK1}	NA
EVT-P-0022936-T01-IM6-98	P-0022936-T01-IM6		STK11-NTRK1	STK11	unknown	five_prime	156848955	15	616	False	disrupted	0.3888888888888889	True			Protein kinase domain	NTRK1 (NM_002529) - STK11 (NM_000455) rearrangement: t(1;19)(;)(chr1:g.56848955::chr19:g.204186) | Protein Fusion: mid-exon {NTRK1:STK11}	NA
EVT-P-0024875-T01-IM6-99	P-0024875-T01-IM6		TAFA2-NTRK1	TAFA2	unknown	three_prime	156836548	4	120	True	retained	1.0	False	Protein kinase domain			NTRK1 (NM_002529) rearrangement: t(1;12)(q23.1;q14.1)(chr1:g.156836548::chr12:g.62314748) | Transcript Fusion {FAM19A2:NTRK1}	NA
EVT-P-0069757-T01-IM7-100	P-0069757-T01-IM7		TARS2-NTRK1	TARS2	unknown	three_prime	156838028	5	187	False	retained	1.0	False	Protein kinase domain			TARS2 (NM_025150) - NTRK1 (NM_002529) fusion: c.264-188:TARS2_c.561:NTRK1del | Protein Fusion: mid-exon {TARS2:NTRK1}	NA
EVT-P-0050767-T01-IM6-101	P-0050767-T01-IM6		TPM3-NTRK1	TPM3	in-frame	three_prime	156845252	12	452	True	retained	1.0	False	Protein kinase domain			TPM3(NM_153469) - NTRK1 (NM_002529) Fusion(TPM3 exon 12 with NTRK1 exon12): c.665-654:TPM3_c.1355-60:NTRK1inv | Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-0049078-T04-IM6-102	P-0049078-T04-IM6		TPM3-NTRK1	TPM3	unknown	three_prime	156843537	8	321	False	retained	1.0	False	Protein kinase domain			TPM3 (NM_001278190) - NTRK1 (NM_002529) fusion: c.395-6179:TPM3_c.963:NTRK1inv | Protein Fusion: mid-exon {TPM3:NTRK1}	NA

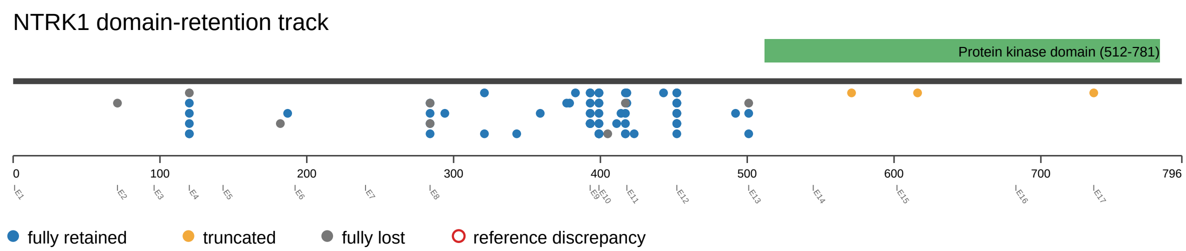
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049078-T01-IM6-103	P-0049078-T01-IM6		TPM3-NTRK1	TPM3	unknown	three_prime	156843537	8	321	False	retained	1.0	False	Protein kinase domain			TPM3 (NM_001278190) - NTRK1 (NM_002529) fusion: c.395-6179:TPM3_c.963:NTRK1inv | Protein Fusion: mid-exon {TPM3:NTRK1}	NA
EVT-P-0046877-T02-IM6-104	P-0046877-T02-IM6		TPM3-NTRK1	TPM3	in-frame	three_prime	156843811	8	393	True	retained	1.0	False	Protein kinase domain			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.776-357:TPM3_c.1177+60:NTRK1inv | Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-0026375-T04-IM6-105	P-0026375-T04-IM6		TPM3-NTRK1	TPM3	in-frame	three_prime	156844046	9	393	True	retained	1.0	False	Protein kinase domain			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.775+355:TPM3_c.1178-129:NTRK1inv | Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-0026375-T01-IM6-106	P-0026375-T01-IM6		TPM3-NTRK1	TPM3	in-frame	three_prime	156844046	9	393	True	retained	1.0	False	Protein kinase domain			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 9-17): c.775+355:TPM3_c.1178-129:NTRK1inv | Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-0025815-T02-IM6-107	P-0025815-T02-IM6		TPM3-NTRK1	TPM3	in-frame	three_prime	156844230	9	399	True	retained	1.0	False	Protein kinase domain			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-6 with NTRK1 exons 10-17): c.*6032:TPM3_c.1195+38:NTRK1inv | Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-0021467-T01-IM6-108	P-0021467-T01-IM6		TPM3-NTRK1	TPM3	in-frame	three_prime	156844038	9	393	True	retained	1.0	False	Protein kinase domain			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-6 fused in-frame to NTRK1 exons 9-17): c.*652:TPM3_c.1178-137:NTRK1inv | Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-0021170-T03-IM6-109	P-0021170-T03-IM6		TPM3-NTRK1	TPM3	in-frame	three_prime	156845218	12	452	True	retained	1.0	False	Protein kinase domain			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 12-17): c.775+235:TPM3_c.1355-94:NTRK1inv | Protein Fusion: in frame {TPM3:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021170-T02-IM6-110	P-0021170-T02-IM6		TPM3-NTRK1	TPM3	in-frame	three_prime	156845218	12	452	True	retained	1.0	False	Protein kinase domain			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 12-17): c.775+235:TPM3_c.1355-94:NTRK1inv | Protein Fusion: in frame {TPM3:NTRK1}	NA
EVT-P-0014143-T01-IM5-111	P-0014143-T01-IM5		TPM3-NTRK1	TPM3	in-frame	three_prime	156844002	9	393	True	retained	1.0	False	Protein kinase domain			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-10 fused in-frame to NTRK1 exons 9-17): c.*1568:TPM3_c.1178-173:NTRK1inv | Protein fusion: in frame (TPM3-NTRK1)	NA
EVT-P-0069266-T01-IM7-112	P-0069266-T01-IM7		TPM3-NTRK1	TPM3	unknown	three_prime	156843601	8	343	False	retained	1.0	False	Protein kinase domain			TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.854+282:TPM3_c.1027:NTRK1inv | Protein Fusion: mid-exon {TPM3:NTRK1}	NA
EVT-P-0049951-T01-IM6-113	P-0049951-T01-IM6		TPR-NTRK1	TPR	in-frame	three_prime	156844660	11	418	True	retained	1.0	False	Protein kinase domain			TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2610+187:TPR_c.1252-38:NTRK1inv | Protein Fusion: in frame {TPR:NTRK1}	NA
EVT-P-0042965-T01-IM6-114	P-0042965-T01-IM6		TPR-NTRK1	TPR	unknown	three_prime	156843703	8	377	False	retained	1.0	False	Protein kinase domain			TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+531:TPR_c.1129:NTRK1inv | Protein Fusion: mid-exon {TPR:NTRK1}	NA
EVT-P-0065034-T02-IM7-115	P-0065034-T02-IM7		TPR-NTRK1	TPR	unknown	three_prime	156843651	8	359	False	retained	1.0	False	Protein kinase domain			TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2777-512:TPR_c.1077:NTRK1inv | Protein Fusion: mid-exon {TPR:NTRK1}	NA
EVT-P-0016347-T01-IM6-116	P-0016347-T01-IM6		TRIM63-NTRK1	TRIM63	in-frame	three_prime	156844048	9	393	True	retained	1.0	False	Protein kinase domain			TRIM63 (NM_032588) - NTRK1 (NM_002529) Rearrangement : c.1051+386:TRIM63_c.1178-127:NTRK1inv | Protein Fusion: in frame {TRIM63:NTRK1}	NA
EVT-P-0007755-T02-IM5-117	P-0007755-T02-IM5		TRPM8-NTRK1	TRPM8	out-of-frame	three_prime	156845664	12	501	True	retained	1.0	False	Protein kinase domain			TRPM8 (NM_024080) - NTRK1 (NM_002529) rearrangement: t(1;2)(q23.1; q27.1)(chr1:g.156845664::chr2:g.234866124) | Protein fusion: out of frame (TRPM8-NTRK1)	NA

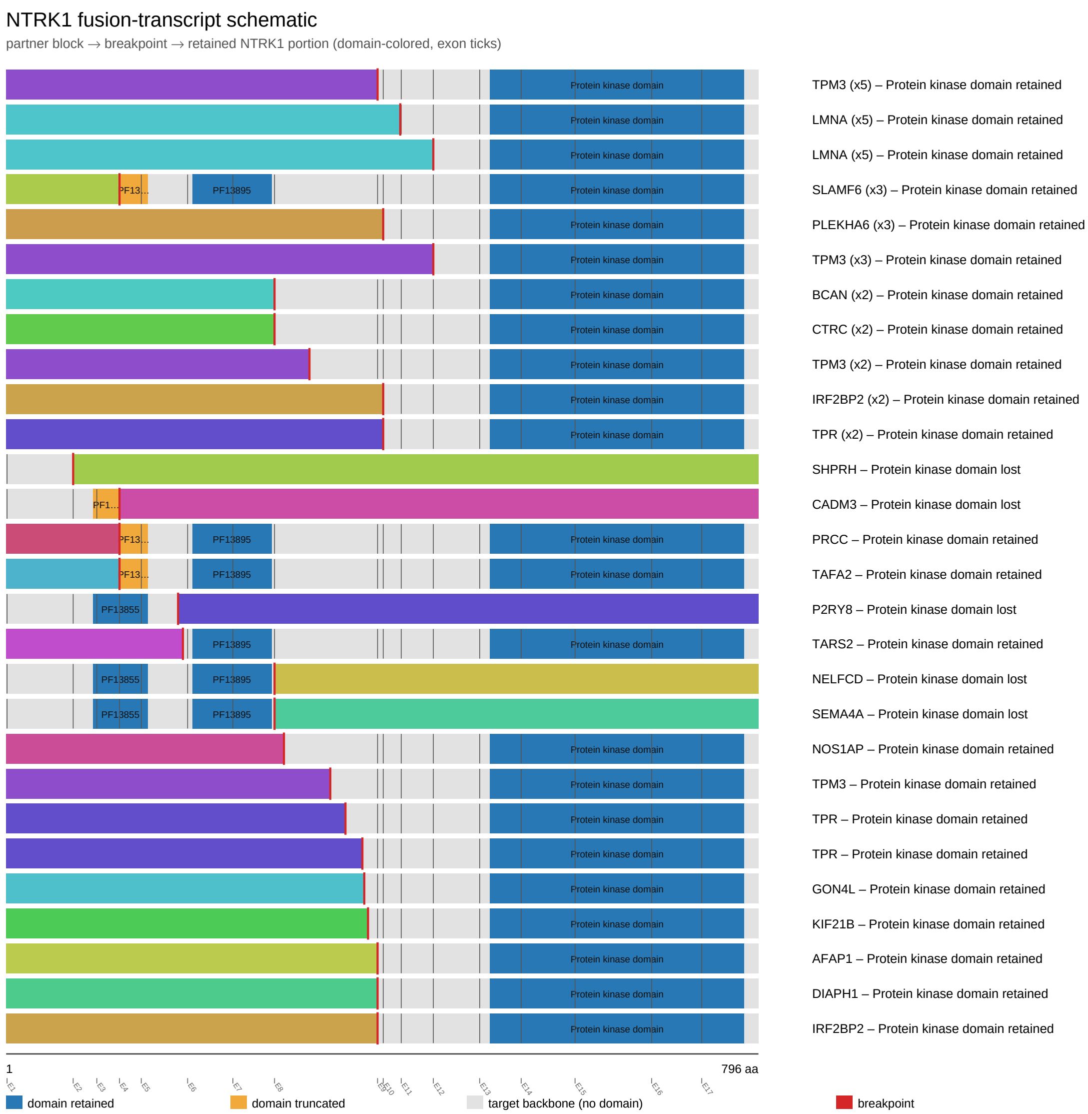
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000104-T02-IM6-118	P-0000104-T02-IM6		ZBTB7B-NTRK1	ZBTB7B	unknown	three_prime	156845173	12	452	True	retained	1.0	False	Protein kinase domain			ZBTB7B (NM_001252406) - NTRK1 (NM_002529) Rearrangement : c.944:ZBTB7B_c.1355-139:NTRK1 del | Protein fusion: mid-exon (ZBTB7B-NTRK1)	NA

Figures

domain retention outliers



fusion schematic

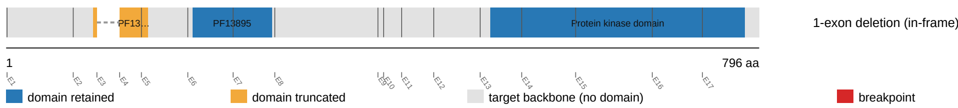


Showing the top 28 of 53 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic

NTRK1 intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



reference comparison

Reference vs msk_impact_50k_2026